

Jack symmetric polynomials and their connection to the Lie algebra of infinitesimal conformal transformations

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Outline

- 1 The Virasoro algebra
- 2 Symmetric polynomials
- 3 Singular vectors of the Virasoro algebra

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The Virasoro algebra

Central extension of the Lie algebra of infinitesimal conformal transformations.

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- Generators and the Lie bracket:

$$\mathcal{L} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} L_n \oplus \mathbb{C} c$$

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}m(m^2 - 1)\delta_{m, -n},$$

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- Triangular decomposition:

$$\mathcal{L}_{\pm} = \bigoplus_{n \geq 1} \mathbb{C} L_{\pm n}, \quad \mathcal{L}_0 = \mathbb{C} L_0 \oplus \mathbb{C} c,$$

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- Verma modules $M(h), h \in \mathbb{C}$:

$$M(h) = U(\mathcal{L}) \otimes_{U(\mathcal{L}_{\geq 0})} \mathbb{C} u_h,$$
$$L_0 u_h = h u_h, \quad L_n u_h = 0, n \geq 1,$$
$$M(h) \cong \bigoplus_{\lambda} \mathbb{C} L_{-\lambda_1} L_{-\lambda_2} \cdots u_h, \quad \lambda \in \text{partitions of integers}$$

These Verma modules are irreducible for generic h and reducible only if h and c satisfy certain polynomial relations [Kac].

Reducible Verma modules

$M(h)$ reducible

There exists a *singular vector* $\chi \in M(h)$ such that

$$\begin{aligned} L_n \chi &= 0, \quad n \geq 1, \\ L_0 \chi &= h' \chi, \quad h' - h \in \mathbb{N}. \end{aligned}$$

χ generates a sub-Verma module $M(h')$ in $M(h)$.

- Singular vectors of crucial importance to conformal field theory.
- Explicit formulae for h' in terms of h and c known [Kac, Feigin, Fuchs]
- No general formulae known for χ

The Heisenberg algebra

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- The Heisenberg algebra:

$$\widehat{\mathfrak{gl}}(1) = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} a_n \oplus \mathbb{C} 1,$$
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- Verma modules $F(\mu), \mu \in \mathbb{C}$ (also called Fock spaces):

$$F(\mu) = U(\widehat{\mathfrak{gl}}(1)) \otimes_{U(\widehat{\mathfrak{gl}}(1)_{\geq 0})} \mathbb{C} |\mu\rangle,$$
$$a_0 |\mu\rangle = \mu |\mu\rangle, \quad a_n |\mu\rangle = 0, n \geq 1,$$
$$F(\mu) \cong \bigoplus_{\lambda} \mathbb{C} a_{-\lambda_1} a_{-\lambda_2} \cdots |\mu\rangle, \quad \lambda \in \text{partitions of integers}$$
$$\cong \mathbb{C}[a_{-1}, a_{-2}, \dots]$$

Fock spaces are simple for all values of $\mu \in \mathbb{C}$.

Finding the Virasoro algebra in the Heisenberg algebra

Formulae for Virasoro generators, $\alpha \in \mathbb{C}$

$$L_n = \frac{1}{2} \sum_{m \in \mathbb{Z}} a_m a_{n-m} - \frac{\alpha}{2} (n+1) a_n, \quad n \neq 0$$

$$L_0 = \frac{1}{2} a_0 (a_0 - \alpha) + \sum_{m \geq 1} a_{-m} a_m, \quad c = 1 - 3\alpha^2$$

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Fock spaces also Virasoro modules

$$L_n |\mu\rangle = 0, \quad n \geq 1$$

$$L_0 |\mu\rangle = h_\mu |\mu\rangle, \quad h_\mu = \frac{\mu(\mu - \alpha)}{2}$$

Fock spaces are reducible over the Virasoro algebra whenever the Verma module $M(h_\mu)$ is reducible.

Virasoro singular vectors in Fock spaces

Singular vectors

A vector $\chi \in F(\mu)$ is singular if

$$\begin{aligned} L_n \chi &= 0, \quad n \geq 1, \\ L_0 \chi &= h' \chi, \quad h' - h_\mu \in \mathbb{N}. \end{aligned}$$

- Unlike Virasoro Verma modules, reducible Fock spaces need not contain singular vectors.
- When they do, elegant formulae can be derived because the Heisenberg algebra is so simple.
- This turns out to be sufficient for conformal field theory purposes.

How to find singular vector formulae

- If there exists a Virasoro homomorphism $\phi : F(\mu) \rightarrow F(\nu)$ so that $\chi = \phi(|\mu\rangle) \neq 0$, then χ is a singular vector in $F(\nu)$

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- Such homomorphisms are given by *screening operators*
[Dotsenko, Fateev]: For all $P \in \mathbb{C}[a_{-1}, a_{-2}, \dots]$

$$P|\mu\rangle \mapsto \int_C \prod_{1 \leq i \neq j \leq r} (1 - \frac{z_i}{z_j})^{1/\kappa} \prod_{i=1}^r z_i^{-s} \prod_{m \geq 1} \exp \left(\lambda \frac{p_m(z)}{m} a_{-m} \right) \\ \times \prod_{m \geq 1} \exp \left(-\lambda \frac{p_m(z^{-1})}{m} a_m \right) P|\nu\rangle \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r},$$

where $\kappa, \lambda \in \mathbb{C}$ depend on α ; $s \in \mathbb{Z}$; μ, ν depend on r, s and α ; and the p_m are power sums

$$p_m(z) = \sum_{i=1}^r z_i^m, \quad p_m(z^{-1}) = \sum_{i=1}^r z_i^{-m}.$$

How to find singular vector formulae

- The images of the screening operators are therefore vectors of the form

$$\int_C \prod_{1 \leq i \neq j \leq r} (1 - \frac{z_i}{z_j})^{1/\kappa} f(z^{-1}) g(z) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r} Q|v\rangle$$

for some symmetric polynomials f, g and for some $Q \in \mathbb{C}[a_{-1}, a_{-2}, \dots]$.

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for some symmetric polynomials f, g and for some $Q \in \mathbb{C}[a_{-1}, a_{-2}, \dots]$.

- These integrals can be evaluated explicitly, because they define a known inner product $\langle f, g \rangle_r^\kappa$ on the ring of symmetric polynomials.

$$\langle f, g \rangle_r^\kappa = \int_C \prod_{1 \leq i \neq j \leq r} (1 - \frac{z_i}{z_j})^{1/\kappa} f(z^{-1}) g(z) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

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Symmetric polynomials

- One can consider symmetric polynomials in either a finite or infinite number of variables:

- ▶ $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$

- ▶ $\Lambda = \varprojlim_r \Lambda_r$

Λ_r can be obtained from Λ or $\Lambda_{r'}, r' > r$ by setting $z_i = 0$ for $i > r$.

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 - ▶ $\Lambda = \varprojlim_r \Lambda_r$
- Λ_r can be obtained from Λ or $\Lambda_{r'}, r' > r$ by setting $z_i = 0$ for $i > r$.
- Both Λ and Λ_r are naturally graded by the degree of polynomials

$$\Lambda_r = \bigoplus_{m \geq 0} \Lambda_r^m, \quad \Lambda = \bigoplus_{m \geq 0} \Lambda^m.$$

Dimensions are

$$\dim \Lambda^m = \#\{\text{partitions of } m\}$$

$$\dim \Lambda_r^m = \#\{\text{partitions of } m \text{ of length at most } r\}.$$

Thus $\dim \Lambda^m = \dim \Lambda_r^m$ if $r \geq m$.

Symmetric polynomials

The rings Λ_r, Λ admit many bases and generators:

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- The elementary symmetric polynomials

$$e_m = \sum_{1 \leq i_1 < \dots < i_m} z_{i_1} \cdots z_{i_m}$$

are also generators:

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Symmetric polynomials

The symmetric monomials form a basis of Λ_r and Λ .

$$m_\lambda = \sum_{\sigma} z_{\sigma(1)}^{\lambda_1} z_{\sigma(2)}^{\lambda_2} \cdots,$$

where σ runs over all distinct permutations of λ . For example

- $m_{[2,1]}(z_1, z_2) = z_1^2 z_2 + z_1 z_2^2$
- $m_{[2,1]}(z_1, z_2, z_3) = z_1^2 z_2 + z_1 z_2^2 + z_1^2 z_3 + z_1 z_3^2 + z_2^2 z_3 + z_2 z_3^2$
- $m_{[s^r]}(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s$
- if $\ell(\lambda) > r$, then $m_\lambda = 0$ in Λ_r

Jack symmetric polynomials

Jack symmetric polynomials P_{λ}^{κ} are characterised by two properties:

- They are upper triangular in symmetric monomials with respect to the dominance ordering of partitions:

$$P_{\lambda}^{\kappa} = \sum_{\mu \leq \lambda} u_{\mu, \lambda}(\kappa) m_{\mu}, \quad u_{\lambda, \lambda}(\kappa) = 1,$$

where $\mu \leq \lambda$ if $|\mu| = |\lambda|$ and

$$\sum_{i=1}^n \mu_i \leq \sum_{i=1}^n \lambda_i, \quad \forall n \geq 1.$$

- Jack polynomials are mutually orthogonal with respect to the inner product $\langle \cdot, \cdot \rangle_r^{\kappa}$:

$$\langle P_{\lambda}^{\kappa}, P_{\mu}^{\kappa} \rangle_r^{\kappa} = 0 \quad \text{if } \lambda \neq \mu.$$

Jack symmetric polynomials

The dominance ordering of partitions is only partial, not total. Constructing the Jack polynomials by Gram-Schmidt orthogonalisation is therefore overdetermined. Yet it works (with a few restrictions on κ).

Important properties of Jack polynomials:

- The “norm square” $\langle P_\lambda^\kappa, P_\lambda^\kappa \rangle_r^\kappa$ is given by known combinatorial formulae.
- Jack polynomials satisfy the identity

$$\prod_{i,j \geq 1} (1 - y_i z_j)^{-1/\kappa} = \prod_{m \geq 1} \exp \left(\frac{1}{\kappa} \frac{p_m(y) p_m(z)}{m} \right) = \sum_{\lambda} b_{\lambda}(\kappa) P_{\lambda}^{\kappa}(y) P_{\lambda}^{\kappa}(z),$$

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- In Λ_r

$$P_{[s^r]}^{\kappa}(z_1, \dots, z_r) = m_{[s^r]}(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s, \quad \forall \kappa.$$

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The strategy for finding singular vector formulae

- The Heisenberg generators $a_{-n}, n \geq 1$ commute. Thus, $U(\widehat{\mathfrak{gl}}(1)_-) = \mathbb{C}[a_{-1}, a_{-2}, \dots]$.
- Power sums generate Λ . Thus, $\Lambda = \mathbb{C}[p_1, p_2, \dots]$.
- The secret to finding singular vector formulae lies in identifying these two polynomial algebras appropriately.

- Recall the screening operators from before:

$$P|\mu\rangle \mapsto \int_C \prod_{1 \leq i \neq j \leq r} (1 - \frac{z_i}{z_j})^{1/\kappa} \prod_{i=1}^r z_i^{-s} \prod_{m \geq 1} \exp\left(\lambda \frac{p_m(z)}{m} a_{-m}\right) \\ \times \prod_{m \geq 1} \exp\left(-\lambda \frac{p_m(z^{-1})}{m} a_m\right) P|\nu\rangle \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r},$$

- Let $\rho : \Lambda \rightarrow U(\widehat{\mathfrak{gl}}(1)_-)$ be the ring homomorphism characterised by $\rho(p_m(y)) = \lambda \kappa a_{-m}$. Then,

$$\rho\left(\prod_{m \geq 1} \exp\left(\frac{1}{\kappa} \frac{p_m(y)p_m(z)}{m}\right)\right) = \prod_{m \geq 1} \exp\left(\lambda \frac{p_m(z)}{m} a_{-m}\right)$$

Singular vectors [Mimachi, Yamada], [Tsuchiya, SW]

Combining what we have learned, we can compute the image of $|\mu\rangle$ in $F(\mathbf{v})$

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 &= \text{const} \cdot \rho(P_{[s^r]}^{\kappa}(y)) |\mathbf{v}\rangle.
 \end{aligned}$$

Conclusion and outlook

- Fock space singular vectors have simple expressions in terms of Jack polynomials with rectangular partitions.
- These methods were used to drastically shorten and simplify old proofs of the classification of the simple modules of the minimal model conformal field theories. [Ridout, SW]
- This method was a crucial tool in the classification of the simple modules of certain logarithmic extensions of the minimal models. [Tsuchiya, SW]
- Essentially the same method also works for free field realisations of $\widehat{\mathfrak{sl}}(2)$. [Ridout, SW; coming soon]
- The same methods should also work for higher rank affine Lie algebras and Lie superalgebras.

The End

Thank you for your attention!